

Development Plan

Software Engineering

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Table 1: Revision History

Date	Developer(s)	Change
Date1	Name(s)	Description of changes
Date2	Name(s)	Description of changes
...	...	...

1 Introduction

This document outlines the proposed development plan for the team. Sections 1-3 contain legal and regulatory information, sections 4-6 contain logistical details for team communication and roles, and Section 7-10 contain timeline and technology logistics. The team charter is also included at the end of this document.

2 Confidential Information?

The images used to train the machine learning model are confidential and cannot be shared publicly. The images will be provided by various municipalities and are subject to a confidentiality agreement. The team will not share any images or data that is considered confidential. A confidentiality agreement has not been completed yet, but a template is available via this link: [Capstone Template](#).

3 IP to Protect

For this project, there is IP to protect. An IP agreement has not been completed yet, but a template is available via this link: [Capstone Template](#).

4 Copyright License

The team is explicitly choosing not to include a copyright license for this project, meaning that the code is under exclusive copyright. This information is also available in the license file: [LICENSE](#).

5 Team Meeting Plan

Team members will meet on a weekly basis on Tuesdays from 11 AM - 12:30 PM. Meetings will be located in-person at one of McMaster's libraries. Team members are expected to come prepared by having reviewed the current state of the deliverable at that time. Meetings may occur more than once per week if needed, which will be held virtually on Discord. Supervisor meetings will be held regularly on a biweekly basis, but the team will be open to meeting more frequently if the supervisor requests it. All members will take turns chairing meetings and taking meeting notes. Meetings should be structured as follows.

At the start of every meeting, the chair will go over the purpose of the meeting and discuss what is on the agenda. The meeting notetaker should take note of the agenda.

After outlining the agenda, the chair will go through each meeting item one-by-one. Each item should have a person responsible for that task to report on

progress, expected outcomes, and explain next steps if necessary. Each item should have a clear time allocation, and expected deadline. The meeting notetaker should take note of all of this information.

At the end of the meeting, the chair will summarize all key decisions made, and restates actions items for the week. The meeting notetaker will write a section for meeting summary notes.

6 Team Communication Plan

Team will use the following communication methods:

- Issues: GitHub Issues
- Regular Weekly Meetings: In-person at McMaster Libraries
- Virtual Meetings: Discord
- Stakeholder Meetings: Teams
- Meeting minutes & Agenda: GitHub
- Project discussion: Discord
- Urgent/Emergency Communication: SMS Messaging

Team members will regularly check in on communication channels for updates, and enable notifications. Team members should make an effort to respond to messages within 24 hours. Team members will exchange phone numbers to communicate in case of emergency or urgent situations.

7 Team Member Roles

- Note Taker: Takes notes to document key events that occurred during weekly team meetings, summarizing key decisions made during the meeting. Responsible for producing organized, neat, and concise notes made accessible to the team for reference.
- Meeting Chair: Is responsible for leading discussion for weekly team meetings. Should outline a meeting agenda in advance to facilitate smooth discussion of current issues and action items.
- Task Owner: Responsible for complete ownership of a particular task or deliverable. During a team meeting, they are expected to know important details regarding a task's progress and expected deadlines. They should be prepared to report on a task during weekly stand-up.

- LLaisson: Responsible for external communication with faculty and all 3rd parties.

8 Workflow Plan

Git will be used for version control and collaboration. The team will use GitHub to host the repository and manage issues, pull requests, and code reviews. Each team member will be responsible for creating their own branches for their work and submitting pull requests for review before merging into the main branch. A general workflow plan is listed below.

- When a deliverable or task is necessary, an issue will be created in the GitHub repository. The issue will contain a description of the task, expected outcomes, and any relevant information. The issue will be assigned to a team member who will be responsible for completing the task.
- The team member will create a new branch for the task and work on it. The branch should be named according to the issue number and a brief description of the task.
- Once the task is completed, the team member will submit a pull request for review. The pull request should contain a description of the changes made, any relevant information, and a link to the issue.
- The team member will request a review from at least one other team member. The reviewer will review the code, provide feedback, and approve or request changes to the pull request.
- Once the pull request is approved, the team member will merge the changes into the main branch. The issue will be closed, and the task will be considered complete.

GitHub Issues will be used to track deliverables/tasks, lectures, and supervisor/TA/team meetings. For deliverables and tasks, tags will be used to identify the kind of task that needs to be done. This could include tags like "bug fixing", "feature development", "documentation", etc. For meetings, tags will be used to identify the type of meeting.

GitHub Actions will be used as the CI/CD platform, and will be used for linting, unit testing, and compiling any documentation written in LaTeX. These workflows will be automatically on push and pull requests, and ensure that both code and documentation are up to standard.

9 Project Decomposition and Scheduling

The team will use Github Projects to track tasks and deliverables. The team will use a Kanban board to track the progress of tasks and deliverables. The board

will have columns for "To Do", "In Progress", "In Review", and "Done". Each task or deliverable will be represented by a card on the board, which will contain a description of the task, expected outcomes, and any relevant information. The cards will be moved across the columns as the task progresses. Each card will be tied to a GitHub issue, which will be used to track the progress of the task. The issue will be assigned to a team member who will be responsible for completing the task.

The project schedule below follows the timeline detailed in the course syllabus.

- Weeks 4: Problem Statement, Proof of Concept (POC) Plan, Development Plan, Team Charter
- Week 6: Requirements Document (SRS) Revision 0, Hazard Analysis Revision 0
- Week 8: Verification & Validation (V&V) Plan Revision 0
- Week 10: Design Document Rev -1 and CD-Fall Rev 0
- Weeks 11+12: Proof of Concept Demo + Interviews
- Week 16: Design Document Revision 0
- Week 18+19: Revision 0 Demo + Interviews
- Week 22: V&V Report Rev 0 and CD-Winter Rev 0
- Week 24: Final Demonstration (Revision 0)
- Week 26: EXPO Demonstration, Final Documentation (Revision 1)

10 Proof of Concept Demonstration Plan

The main risk towards the success of the project is the ability to accurately detect and classify the state of the road based on the information given. During the proof of concept demonstration, we will demonstrate that our model can be given an image of a road and accurately detect and classify the state of the road.

Some major risks and corresponding mitigation strategies are as follows:

- Risks associated with the quality and quantity of images provided by municipalities. Currently, the project assumes that the team will have a large, labelled dataset of images to train the model with.
 - To address this, the team will do research into public datasets that can be used to train the model. The team will also investigate data augmentation techniques to increase the size of the dataset.

- Another major risk is the model’s ability to classify the state of the road in a way that is universally applicable to all municipalities. The model may be able to classify the state of the road in a way that is specific to a certain municipality, but not applicable to others.
 - As a midway point, the model will be trained to classify the state of the road in a way that is applicable to a specific municipality. The team will then investigate how to generalize the model to work for other municipalities.

11 Expected Technology

Please note that the technologies and tools listed above are not final and may change as the project progresses. The team will evaluate the effectiveness of the tools and technologies used and make adjustments as necessary.

Python will be the primary programming language used for this project. We will be using libraries such as OpenCV for image processing, TensorFlow or PyTorch for machine learning, and NumPy for numerical computations. We will also utilize pre-trained models for road state detection and classification.

Ruff will be used as the linter for Python code, and PyTest will be used as the unit testing framework. We will investigate code coverage measuring tools such as Coverage.py to ensure that our tests are comprehensive.

For version control, we will use Git and GitHub. We will use CI for linting and unit test passing. Since we are using a Github repo, Github Actions is the preferred method of choice for CI.

12 Use of AI and LLMs

AI and LLMs will be used to assist in the development of code and debugging. The team will also use AI to assist in the writing of documentation. The agents that the team will use are Github Copilot, Claude Sonnet 5.5, and Google Gemini Flash 3.6. All code and documentation generated by AI will be reviewed by at least two members of the team to ensure accuracy and quality, and must pass linting and unit tests before being pushed into a branch. Additionally, planning mode will be used for any agents that are used to ensure that the output is accurate and relevant to the task at hand.

13 Coding Standard

PEP 8 will be used as the coding standard for Python code. In the event that a different language is used, the team will abide by the standards set by the creators of the language.

In the event that the coding standard fails to cover a specific case, the team will discuss and agree on a standard to follow for that specific case. Linters will be used to ensure consistent and clean code according to the specified standards.

14 Appendix — Generative AI Use Disclosure

14.1 AI Tools Used

Github Copilot, version 1.0.88.

14.2 How and Where Used

Github Copilot was used to complete sentences in the use of AI + LLM, expected technology, and workflow plan sections.

14.3 Verification of AI-Generated Content

Since Copilot only completes sentences, the team wrote short, concise sentences that were then expanded by Copilot. This way, any ideas in the document are the team's own, and Copilot is only used to expand on those ideas. For any documentation generated by Copilot, a second member of the team reviewed the content before allowing it to be pushed. For any technologies that were suggested by Copilot, the team manually researched the technology to ensure that it was relevant and applicable to the project.

Appendix — Reflection

The purpose of reflection questions is to give you a chance to assess your own learning and that of your group as a whole, and to find ways to improve in the future. Reflection is an important part of the learning process. Reflection is also an essential component of a successful software development process.

Reflections are most interesting and useful when they're honest, even if the stories they tell are imperfect. You will be marked based on your depth of thought and analysis, and not based on the content of the reflections themselves. Thus, for full marks we encourage you to answer openly and honestly and to avoid simply writing "what you think the evaluator wants to hear."

Please answer the following questions. Some questions can be answered on the team level, but where appropriate, each team member should write their own response:

1. Why is it important to create a development plan prior to starting the project?
 - It establishes a clear structure for how the team will develop the project.
 - It ensures that all team members understand their roles and responsibilities.
 - It establishes consistent communication and meeting expectations.
 - It defines the team's workflow for issues, branches, pull requests, and reviews.
 - It helps the team identify project risks and mitigation strategies early.
 - It provides a schedule that allows the team to track progress throughout the project.
2. In your opinion, what are the advantages and disadvantages of using CI/CD?
 - CI/CD automatically checks changes before they are integrated into the project.
 - It helps identify linting, testing, and documentation errors early.
 - It helps maintain consistent code and documentation quality across the team.
 - It reduces the amount of repetitive testing that team members must perform manually.
 - Setting up and maintaining CI/CD workflows requires additional development time.
 - Incorrectly configured workflows can cause valid changes to fail automated checks.

3. What disagreements did your group have in this deliverable, if any, and how did you resolve them?
- We had minor differences in opinion about which technologies and tools to use.
 - We discussed the advantages and disadvantages of each option as a team.
 - We agreed to keep technologies flexible when further investigation is required.
 - We also discussed how responsibilities should be divided between team members.
 - We resolved these discussions by agreeing on shared roles and rotating responsibilities.
 - Final decisions were made collaboratively after considering each member's input.

Appendix — Team Charter

[borrows from [University of Portland Team Charter](#) —SS]

External Goals

[What are your team’s external goals for this project? These are not the goals related to the functionality or quality of the project. These are the goals on what the team wishes to achieve with the project. Potential goals are to win a prize at the Capstone EXPO, or to have something to talk about in interviews, or to get an A+, etc. —SS]

Attendance

Expectations

[What are your team’s expectations regarding meeting attendance (being on time, leaving early, missing meetings, etc.)? —SS]

Acceptable Excuse

[What constitutes an acceptable excuse for missing a meeting or a deadline? What types of excuses will not be considered acceptable? —SS]

In Case of Emergency

[What process will team members follow if they have an emergency and cannot attend a team meeting or complete their individual work promised for a team deliverable? —SS]

Accountability and Teamwork

Quality

[What are your team’s expectations regarding the quality of team members’ preparation for team meetings and the quality of the deliverables that members bring to the team? —SS]

Attitude

[What are your team’s expectations regarding team members’ ideas, interactions with the team, cooperation, attitudes, and anything else regarding team member contributions? Do you want to introduce a code of conduct? Do you want a conflict resolution plan? Can adopt existing codes of conduct. —SS]

Stay on Track

[What methods will be used to keep the team on track? How will your team ensure that members contribute as expected to the team and that the team performs as expected? How will your team reward members who do well and manage members whose performance is below expectations? What are the consequences for someone not contributing their fair share? —SS]

[You may wish to use the project management metrics collected for the TA and instructor for this. —SS]

[You can set target metrics for attendance, commits, etc. What are the consequences if someone doesn't hit their targets? Do they need to bring the coffee to the next team meeting? Does the team need to make an appointment with their TA, or the instructor? Are there incentives for reaching targets early? —SS]

Team Building

[How will you build team cohesion (fun time, group rituals, etc.)? —SS]

Decision Making

[How will you make decisions in your group? Consensus? Vote? How will you handle disagreements? —SS]